

MOGOCHA, Russian Federation

WMO: 306730

Lat: 53.750N

Lon: 119.733E

Elev: 625

StdP: 94.04

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.5	-38.5	-42.5	0.1	-40.1	-40.5	0.1	-38.3	8.3	-20.9	7.0	-19.9	0.1	140	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	28.4	18.3	26.5	17.5	24.8	16.8	20.0	25.2	18.9	24.0	17.9	22.7	2.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.2	22.2	17.2	13.3	21.1	16.2	12.4	20.1	60.6	24.8	56.7	24.0	53.4	22.7	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.3	5.3	DB	-43.1	31.4	2.5	1.6	-44.9	32.6	-46.3	33.5	-47.7	34.4	-49.5	35.5	(4)
(5)			WB	-43.1	21.9	2.4	1.2	-44.9	22.7	-46.3	23.4	-47.6	24.1	-49.4	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.9	-27.0	-22.0	-12.5	-0.9	7.7	14.8	17.5	14.7	7.1	-3.2	-17.6	-26.6	(6)	
	DBStd	16.69	4.88	5.37	6.46	4.69	3.84	3.38	2.68	3.27	4.09	5.22	7.09	5.18	(7)	
	HDD10.0	5648	1147	897	697	327	92	4	1	4	107	409	829	1135	(8)	
	HDD18.3	8150	1405	1131	955	576	330	112	46	117	338	667	1079	1393	(9)	
	CDD10.0	572	0	0	0	0	20	149	233	151	19	0	0	0	(10)	
	CDD18.3	32	0	0	0	0	0	7	20	5	0	0	0	0	(11)	
	CDH23.3	696	0	0	0	1	17	237	326	113	3	0	0	0	(12)	
	CDH26.7	143	0	0	0	0	2	56	71	16	0	0	0	0	(13)	
(14) Wind	WSAvg	1.6	1.0	1.1	1.8	2.4	2.5	1.7	1.5	1.4	1.7	1.6	1.1	0.9	(14)	
(15) Precipitation	PrecAvg	431	4	4	6	17	34	77	114	98	47	15	9	6	(15)	
	PrecMax	640	11	17	17	43	108	185	231	236	100	54	22	22	(16)	
	PrecMin	250	0	0	0	0	1	12	34	19	9	0	0	0	(17)	
	PrecStd	92	3	4	4	11	21	43	50	45	25	10	5	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.5	-1.9	8.2	18.6	25.9	30.8	30.8	28.8	23.3	14.6	2.6	-8.5	(19)	
		MCWB	-13.0	-5.0	2.4	9.1	14.5	18.7	19.1	18.7	14.5	7.4	-0.9	-9.8	(20)	
	2%	DB	-14.1	-5.9	4.5	13.7	22.6	28.2	28.8	26.3	20.5	10.6	-1.4	-13.3	(21)	
		MCWB	-15.1	-8.1	-0.2	6.0	11.8	17.9	18.8	17.6	13.0	5.1	-3.8	-14.1	(22)	
	5%	DB	-16.5	-9.2	1.6	10.8	20.0	26.0	26.9	24.3	17.8	7.5	-4.4	-15.9	(23)	
		MCWB	-17.3	-10.7	-2.1	4.3	10.7	16.4	18.2	17.2	11.5	3.0	-6.0	-16.5	(24)	
	10%	DB	-18.7	-12.0	-1.0	8.2	17.4	24.0	25.1	22.3	15.4	5.0	-7.0	-18.4	(25)	
		MCWB	-19.3	-13.2	-3.8	2.5	9.4	15.6	17.6	16.2	10.3	1.4	-8.3	-19.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.8	-4.9	2.5	9.5	14.6	21.7	21.6	20.5	16.0	7.6	-0.6	-9.9	(27)	
		MCDB	-11.5	-2.1	7.8	17.5	24.1	26.4	27.0	24.9	20.9	13.9	2.5	-8.7	(28)	
	2%	WB	-15.1	-7.9	0.0	6.5	12.7	19.2	20.4	19.2	14.3	5.2	-3.6	-14.1	(29)	
		MCDB	-14.1	-6.0	4.3	13.4	20.7	25.7	25.8	23.9	18.3	10.1	-1.5	-13.4	(30)	
	5%	WB	-17.2	-10.6	-1.9	4.5	11.3	17.6	19.3	18.1	12.6	3.4	-5.8	-16.5	(31)	
		MCDB	-16.5	-9.2	1.3	10.3	18.1	24.0	24.3	22.4	16.7	7.0	-4.4	-15.9	(32)	
	10%	WB	-19.2	-13.1	-3.8	2.8	9.9	16.4	18.4	17.0	10.9	1.6	-8.3	-18.9	(33)	
		MCDB	-18.7	-12.0	-1.0	7.8	16.5	22.5	23.2	21.3	14.7	4.6	-7.1	-18.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	17.5	16.9	13.9	14.3	15.2	13.0	13.0	13.4	12.6	13.3	11.9	(35)	
		MCDBR	13.7	19.1	18.1	19.4	20.8	20.1	17.8	17.4	18.2	16.2	15.0	13.1	(36)	
		MCWBR	12.9	17.2	13.8	12.2	11.2	10.3	8.9	9.5	10.9	11.0	12.5	12.3	(37)	
	5% WB	MCDBR	13.8	19.0	17.3	18.4	18.2	17.1	14.1	13.9	14.7	15.6	14.5	13.0	(38)	
		MCWBR	12.9	17.0	13.4	11.8	10.2	9.3	7.8	8.1	9.2	10.6	12.1	12.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.239	0.265	0.304	0.402	0.491	0.459	0.425	0.391	0.339	0.304	0.263	0.243	(40)	
	taud		2.194	2.180	2.147	2.048	1.904	2.113	2.272	2.348	2.442	2.412	2.250	2.121	(41)	
	Ebn at Noon		740	835	871	818	759	794	819	828	835	788	697	637	(42)	
	Edh at Noon		61	88	115	147	180	148	124	109	87	71	58	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.22	2.35	4.06	5.32	5.28	5.79	5.29	4.50	3.40	2.28	1.37	0.91	(44)	
	RadStd		0.04	0.05	0.12	0.31	0.39	0.38	0.40	0.38	0.26	0.14	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page